NATIONAL GDP GROWTH PREDICTOR

1. What Type of Dataset Is This?

This is a World Development Indicators–style panel dataset (country-wise time series data).

It contains:

- Countries
- Two economic/social indicators
- Data from 2001 to 2025

Each country appears twice — once for each indicator.

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2. What Variables Are Present?

Identification Columns

1. Country Name – Name of the country
   Example: Afghanistan
2. Country Code – 3-letter country code
   Example: AFG
3. Series Name – Indicator name
   Only 2 indicators:
   - GDP per capita (current US$)
   - Individuals using the Internet (% of population)
4. Series Code – Indicator code
   - NY.GDP.PCAP.CD → GDP per capita
   - IT.NET.USER.ZS → Internet usage %

Year Columns (2001–2025)

Each column like:

2001 [YR2001]

2002 [YR2002]

...

2025 [YR2025]

These represent values of the indicator for that year.

Example:

- GDP per capita in 2010
- Internet users % in 2015

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3. Dataset Structure

- Total Rows: 537
- Total Columns: 29
- Countries: ~266
- Indicators per country: 2

So:

266 countries × 2 indicators = ~532 rows

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4. Important Observations

1. Data Type Issue

All year columns are stored as:

object (string)

They should be converted to numeric for analysis.

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2. Missing Values

Missing values are shown as:

".."

Before analysis, you must:

df.replace("..", pd.NA, inplace=True)

Then convert to numeric:

df.iloc[:, 4:] = df.iloc[:, 4:].apply(pd.to_numeric, errors='coerce')

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5. What Can You Do With This Dataset?

Since you're doing ML and economic projects, this dataset is PERFECT for:

- 1. GDP vs Internet Penetration Analysis
  - Does internet usage increase GDP?

- Correlation analysis
- Regression modeling

2. Time Series Trend Analysis

- GDP growth trends
- Digital development trends

3. Country Comparison

- India vs USA
- Developing vs Developed countries

4. Pandemic Impact Study

Check GDP drop in 2020 (COVID effect)

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6. Example Insight (From Sample Data)

For Afghanistan:

- GDP per capita increased from ~138 (2001) to ~522 (2016)
- Internet usage increased from 0.004% to 11%

Shows digital growth over time.

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7. Dataset Type (Important for Viva)

This is:

Panel Data (Cross-sectional + Time Series Data)

Because:

- Multiple countries (cross-sectional)
- Multiple years (time series)